Loan Default Prediction

Comparing Classic ML and Neural Networks under Class Imbalance

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Final Project Group 15 · Home Credit Default Risk (Kaggle)

Problem & Motivation

Task Binary classification — predict loan default ($\texttt{TARGET} \in \{0, 1\}$)

Data 307,511 applicants $\times$ 122 features (Kaggle)

Challenge only **8.07%** default $\Rightarrow$ severe class imbalance

Approach 5 models (3 classic ML + 2 NN) + imbalance handling on best NN

Metric ROC AUC (accuracy is misleading on imbalanced data)

Research question: do classic ML and NN respond differently to imbalance handling?

Dataset · Home Credit Default Risk

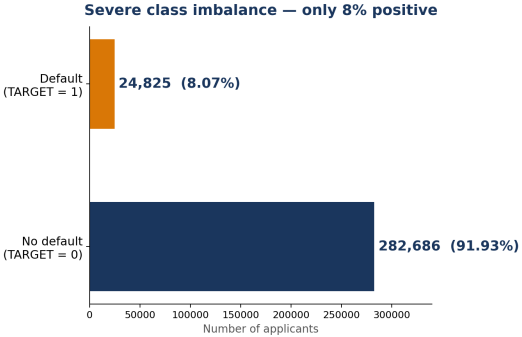
Structure Tabular — row = one applicant

Obs. 307,511 train / 48,744 test

Features 122 (102 numeric, 16 categorical)

Target TARGET $\in \{0, 1\}$; 1 = default

Imbalance **8.07%** default rate



Strongest signals

- EXT_SOURCE_1/2/3 external credit scores
- Demographics, employment, credit bureau

Classic ML Models

Model	Best hyperparameters	Val AUC
Logistic Regression	`C`=0.1; `class_weight=balanced`	0.7487
Random Forest	`n_estimators`=200; `leaf`-regularized; `balanced`	0.7470
HistGradientBoosting	`lr`=0.05; `max_iter`=200; `leaf`=100	**0.7595**

Key takeaways

- HGBC wins by ∼1 AUC point — boosted trees dominate tabular.
- `class_weight='balanced'` on LR / RF handles imbalance in the loss.
- RF overfits at `leaf=1`: train 0.96 vs val 0.747.

Neural Network Models

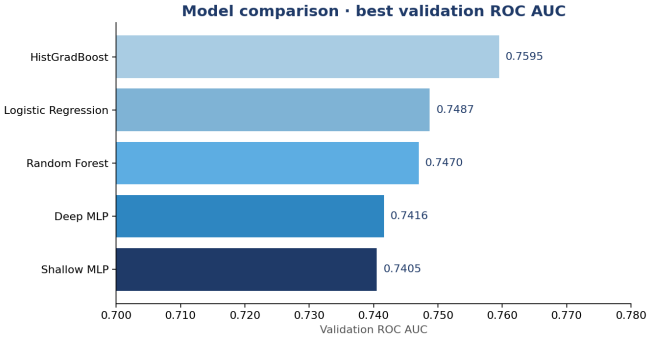
Architecture	Hidden	Parameters	Val AUC
Shallow MLP	$(64,)$	$\sim$15.7K	0.7405
Deep MLP	$(128, 64, 32)$	$\sim$42K	**0.7416**

ReLU · Adam · early stopping.
Tuned: `alpha` $\in \{10^{-4}, 10^{-3}\}$, `lr` $\in \{0.001, 0.005\}$.
Best for both: $\alpha = 10^{-3}$, `lr` $= 0.005$.

Key takeaways

- Depth doesn't help: 0.7405 vs 0.7416 (within noise).
- Tripling parameters bought nothing.
- Both NNs land below the classic learners.

Combined Leaderboard



Model	Val AUC
HistGradBoost	**0.7595**
Logistic Reg.	0.7487
Random Forest	0.7470
Deep MLP	0.7416
Shallow MLP	0.7405

Tree boosting wins by $\sim$1 AUC point.

Imbalance Handling on the Best MLP

Method	Thresh.	AUC	PR-AUC	Prec.	Recall	F1
Baseline	0.500	0.742	0.217	0.00	0.00	0.00
SMOTE	0.500	**0.648**	0.134	0.16	0.21	0.18
Threshold tuning $\star$	0.139	0.742	0.217	0.22	**0.41**	**0.29**

Key takeaways

- Baseline at 0.5: AUC 0.742 but recall **0%**.
- SMOTE hurts: AUC drops 0.09.
- Threshold tuning at 0.139 lifts recall to **41%**.

Threshold > resampling.

Conclusions

Best model overall

HistGradientBoosting takes the top spot at val ROC AUC = **0.7595**, indicating that nonlinear tree boosting captures interactions across heterogeneous features. LR and RF perform similarly, both slightly below HGBC — LR likely benefits from strong linear predictors such as `EXT_SOURCE_*`. Among NNs, Deep MLP barely edges Shallow MLP (0.7416 vs 0.7405); both lag classic shallow learners by $\sim$1–2 AUC pts.

Imbalance handling on the MLP

- Threshold tuning lifts recall **0% $\rightarrow$ 41%**; SMOTE hurts AUC.
- Choosing the threshold matters more than rebalancing the data.

Limitations

- Only main `application_*.csv` tables (auxiliary tables would add several AUC points).
- Single train/val split (PredefinedSplit), not full k-fold CV.

Recording

<<INSERT RECORDING LINK BEFORE SUBMISSION>>

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